

VINOD KUMAR VEMULA

Senior Software Engineer | Agentic AI & Recommendation Systems | Python | AWS | Databricks

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PROFESSIONAL SUMMARY

Senior Software Engineer with 7 years of Python development experience building production-grade AI, ML, and data systems. Deep expertise in agentic AI architecture, REST API engineering, recommendation model development, and backend system design with a strong culture of CI/CD, unit testing, and Git-driven workflows. Hands-on AWS experience (Lambda, S3, Redshift, Glue) and working proficiency with Databricks-style Spark/lakehouse architectures. Comfortable operating in research-and-experiment environments rapidly prototyping, iterating, and shipping pilot AI products. Background in enterprise finance and insurance domains where deep data, backend reliability, and AI-driven next-action recommendations directly drive revenue outcomes. Dallas-area based and available on-site.

TECHNICAL SKILLS

Languages	Python, SQL, R, Java
Agentic / AI	LangGraph (graphs, tool-nodes, memory/state, streaming agents), LangChain, LlamaIndex, MCP (Model Context Protocol), RAG, Prompt Engineering, Context Engineering
Recommendation	Next-best-action models, collaborative filtering, content-based filtering, embedding similarity, real-time scoring APIs, A/B testing frameworks
AWS	Lambda (serverless inference), S3 (data lake), Redshift (analytics DW), Glue (ETL), SageMaker (ML pipelines), CloudFormation, IAM, EventBridge — production experience
Databricks	Lakehouse / Delta Lake architecture, PySpark DataFrames, MLflow experiment tracking, feature stores, large-scale feature engineering (preferred/strong working knowledge)
REST APIs	FastAPI, Pydantic, async endpoints, OpenAPI specs, versioning, JWT auth, rate limiting, integration testing
CI/CD & Testing	GitHub Actions, Jenkins, Azure DevOps, pytest, AsyncMock, unit & integration tests, coverage gates, pre-commit hooks
ML Frameworks	Scikit-learn, XGBoost, TensorFlow, PyTorch, Neural Networks, MLOps, model monitoring & automated retraining
Vector / Search	PGVector, Vertex AI Vector Search, Pinecone, embedding pipelines, semantic similarity
Cloud (Other)	GCP (Vertex AI, BigQuery, Pub/Sub), Azure (ML Studio, Synapse, OpenAI, DevOps)
Open Source	Active GitHub contributor; comfortable adopting, forking, and extending OSS frameworks (LangChain, HuggingFace, FastAPI ecosystem)
IaC & DevOps	Terraform, CloudFormation, Docker, Kubernetes, Cloud Build

PROFESSIONAL EXPERIENCE

Senior Gen AI / Software Engineer — CGI | Dallas, TX, USA

Oct 2025 – Present

- Designed and shipped production agentic AI systems in Python using LangGraph — stateful multi-agent graphs with tool-calling nodes, memory persistence, and streaming outputs reducing manual enterprise decision-processing by 50%; directly analogous to building the Next Action agent suite.
- Engineered REST APIs (FastAPI + Pydantic) exposing AI agent capabilities as backend services: OpenAPI spec, versioning, JWT auth, async endpoints, and rate limiting consumed by downstream business applications with zero downtime deployments.
- Built an MCP server exposing enterprise data and tools as standardized agent context endpoints, enabling autonomous multi-step workflow execution with role-based access control and full audit trails giving AI agents structured, secure access to deep backend data.
- Developed RAG pipelines combining vector retrieval, reranking, and LLM reasoning to power context-aware, next-best-action recommendation outputs reducing hallucinations and improving recommendation relevance in high-volume production environments.
- Maintained full CI/CD pipelines via GitHub Actions: pytest suites, coverage enforcement, integration tests, and automated deployment gates keeping production-quality standards on a rapidly evolving AI codebase.
- Led research and experimentation on new agent memory strategies, LLM reasoning patterns, and retrieval approaches rapidly iterating in a pilot-phase environment and promoting validated patterns to production.

AI / ML Engineer — Senior Python Developer — BMO Bank | Chicago, IL, USA

Jun 2023 – Sep 2025

- Built Python-based recommendation and risk scoring systems (XGBoost, Neural Networks, Azure ML) that surfaced next-best-action product insights to 50+ business stakeholders via backend REST APIs reducing false positives by 25% and improving decision accuracy by 18%.
- Developed real-time inference REST APIs using FastAPI and Azure OpenAI, delivering personalized, AI-generated recommendations based on deep customer transaction history directly mirroring the SGWS Next Action sales recommendation model design.
- Engineered streaming ML inference pipelines for high-volume transaction scoring, applying Kafka-style event-driven patterns for low-latency, continuous model predictions on live data.
- Built and maintained comprehensive pytest suites and CI/CD pipelines (Azure DevOps): unit tests, integration tests, and automated deployment workflows accelerating release cycles by 30% and cutting production errors by 20%.
- Ran structured ML experiments using MLflow-compatible tracking versioning datasets, model artifacts, and hyperparameters in Git-backed workflows in a high-research, high-iteration team environment.
- Performed large-scale feature engineering on Azure Synapse (Spark-based) for ML model training experience directly transferable to Databricks lakehouse and Delta Lake workflows.

Data & AI Systems Engineer — Infosys | Mysore, India

Mar 2019 – Dec 2021

- Built real-time data pipelines in Python on GCP (Pub/Sub, BigQuery, Dataflow), handling 500K+ daily transactions with sub-second latency powering downstream ML models with reliable, clean feature data.
- Developed NLP-based text classification and sentiment analysis services using open-source Python libraries, deployed as REST microservices integrating CI/CD automation (Jenkins, GitHub) to reduce release errors by 15%.
- Managed Git repositories, enforced branching strategies, mandatory code reviews, and automated testing gates across a team of 8 engineers, strong Git workflow discipline from day one.
- Designed GCP data lake architectures (BigQuery, Cloud Storage) supporting ML feature engineering pipelines — Databricks/Spark-transferable data engineering foundations.

Data Engineer — Aviva Life Insurance | Bengaluru, India

Jan 2018 – Feb 2019

- Built backend Python pipelines for real-time insurance transaction processing, improving throughput latency by 30% through optimized event-driven workflow design on AWS (S3, Glue ETL).
- Engineered risk-based ML features transaction patterns, anomaly signals to support fraud detection and underwriting recommendation models, feeding backend AI decision systems used by underwriters.

KEY PROJECTS

Next-Action Product Recommendation System — Personal Research Project

Apr 2026

- Built an agentic recommendation engine in Python using LangGraph that ingests customer purchase history and behavioral signals, runs embedding-based similarity search (PGVector), and surfaces the next most relevant product via a FastAPI backend directly modelling SGWS Next Action suite architecture.
- Implemented A/B-testable recommendation logic with MLflow experiment tracking, full pytest coverage, and GitHub Actions CI/CD demonstrating production discipline on a research-phase pilot prototype.

Real-Time ML Decision System — Personal Project

Mar 2023 – May 2023

- End-to-end Python ML system: feature engineering, model training (Scikit-learn / XGBoost), real-time REST API serving on AWS Lambda + S3, and automated model monitoring with retraining triggers full software engineering lifecycle in a research-first context.

EDUCATION

Master of Science — Applied Statistics & Decision Analytics | Western Illinois University, USA | GPA: 3.60 | Jan 2022 – May 2023

Bachelor of Engineering — Mechanical Engineering | Andhra University, India | GPA: 3.0 | Aug 2014 – May 2018

CERTIFICATIONS

Certified GenAI Expert | **AWS Data Engineer** | **Azure Data Engineer Associate** | **AI & Machine Learning on Google Cloud**